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REVIEW ARTICLE

Recent Advances in Applying Nanobubbles for Enhanced Anaerobic Digestion: Fundamentals, Challenges, and Future Prospects

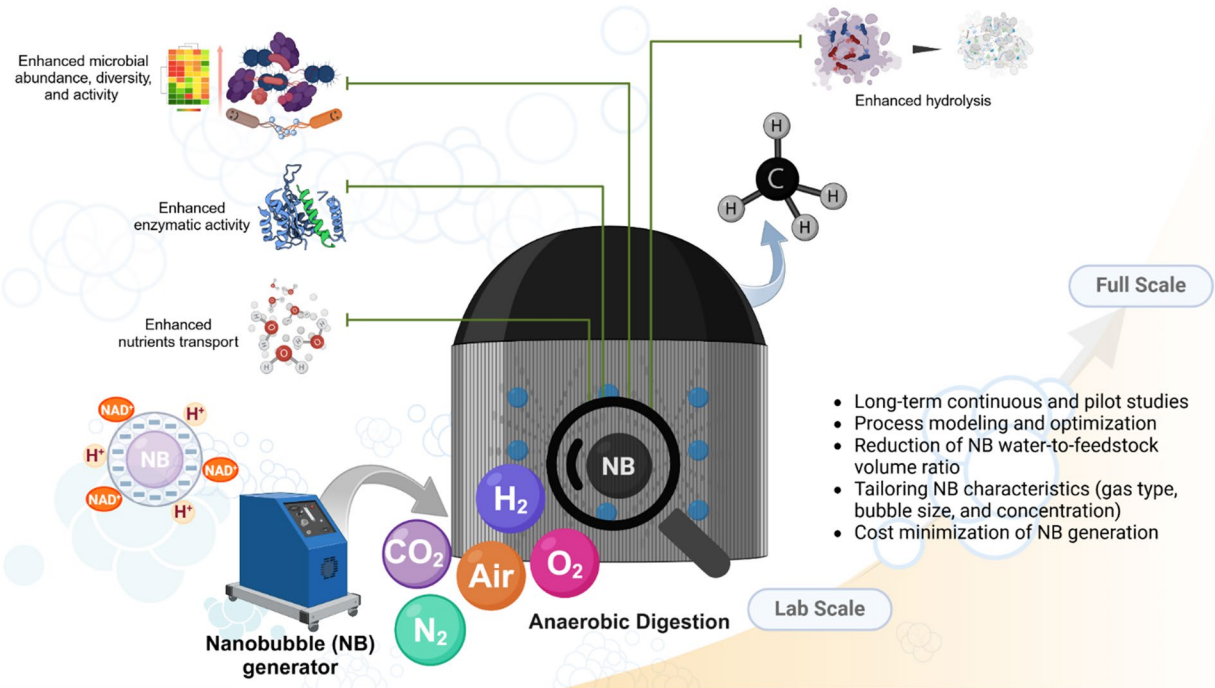
Alsayed Mostafa[1] · Ahmed Elsayed[2] · Rasha Maal-Bared[3] · Elsayed Elbeshbishy[2] · Bipro Ranjan Dhar[1]

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Abstract

In recent years, the use of nanobubbles (NBs) in anaerobic digestion (AD) has received significant research attention, with numerous studies reporting notable improvements in AD efficiency when nanobubble water (NBW) is introduced in digesters. The improvements are often attributed to more efficient substrate degradation, higher activity of key enzymes, and the more complex and diverse microbial communities associated with NBW-integrated AD. Despite these promising results, detailed explanations of the mechanisms and modes of action behind NBW, particularly its impact on each step of the AD process—hydrolysis, acidogenesis, acetogenesis, methanogenesis—are still limited. This paper aims to provide a comprehensive review and critical assessment of NBW's impacts on AD, with a particular focus on how operational conditions can influence NBW-related effects on AD performance. Finally, energy efficiencies, future opportunities and challenges of expanding NB technology to boost AD efficiency are discussed.

Graphical Abstract



Keywords Nano-scale gas bubbles · Biogas production · Anaerobic bioprocess · Gas–liquid mass transfer · Waste-to-energy

Extended author information available on the last page of the article

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Introduction

Given the significant global interest in green energy production, resource recovery, and sustainable practices, anaerobic digestion (AD) has emerged as a promising technology for addressing various challenges, including the management of organic waste and associated greenhouse gas emissions. The AD process comprises four successive steps, i.e., hydrolysis, acidogenesis, acetogenesis and methanogenesis [1, 2]. By treating organic wastes, the AD not only minimizes waste but also produces renewable biogas, which contributes to carbon neutrality and serves as a sustainable alternative to fossil fuels [3]. Also, AD enables nutrient recovery, minimizes energy demand and offers economic and environmental benefits [4]. Despite its multiple advantages, AD process still faces some challenges in terms of instability at high organic loading rates [5]. Challenges also include accumulation of volatile fatty acids (VFAs) when using easily biodegradable feedstocks [2], slow hydrolysis of complex substrates [6], production of inhibitory products such as hydrogen sulfide (H_2S)/bisulfide ion (HS^-) and ammonia (NH_3) [7, 8], and ultimately the inhibition of microbial community leading to process instability [9, 10]. Various sludge pre-treatment approaches, including thermal hydrolysis, chemical oxidation, alkaline treatment, and ultrasonic disintegration, have been employed to address these drawbacks. However, these approaches often involve high chemical/energy consumption, high capital investment, and may produce additional inhibitory by-products [11–13].

An innovative approach used to overcome some of these shortcomings is microaeration, which involves introducing a small amount of oxygen (O_2) or air into digesters [14]. The required gas dosage for maintaining microaerobic conditions has been reported as 0.005 ~ 5 L O_2/$L_{reactor}$/day [15], or maintaining dissolved oxygen (DO) levels between 0.1 ~ 1 mg/L [16]. Microaeration intervals reported in literature vary widely, ranging from intermittent [17] to continuous injection of air or oxygen [18, 19]. Microaeration enhances the abundance of facultative bacteria, as well as acidogenic and hydrolytic microbes, in digesters, thereby improving substrate degradation [20]. Nevertheless, safety concerns, particularly the risk of explosion due to excess oxygen, remain a significant barrier to the widespread implementation in full-scale AD facilities [21]. Unlike conventional microaeration, which involves introducing microbubbles of air or oxygen into digesters, nanobubble (NB) technology involves introducing the gas in nano-sized bubbles, providing better distribution and utilization efficiencies [22].

The concept of NBs was first introduced in 1994 to explain attractive forces observed between hydrophobic surfaces in water [23]. The first image of nano-sized bubbles was captured using atomic force microscopy (AFM) in 2000 [24]. Over the last two decades, NB technology has been applied in biomedical engineering [25], agriculture [26], and environmental remediation [27]. Recently, the application of this technology in AD systems has attracted growing interest [28, 29]. Supplementing NBs in AD enriches hydrolytic bacteria and methanogens, improving hydrolysis and methanogenesis [30, 31]. Further, NBs promote the generation of reactive oxygen species (ROS), which at optimal levels enhance microbial metabolism [30]. In addition, NBs possess unique physicochemical properties, such as high surface charge, large surface area-to-volume ratio, high negative zeta potential, long residence time, weak floating features, low buoyancy, high gas transfer efficiency and advanced gas solubility [22, 32], making them a promising option for enhancing the performance of the AD process [10, 33].

Typically, the NBs are introduced into anaerobic digesters by first producing nanobubble water (NBW), which usually contains approximately 10^6–10^8 gas bubbles per milliliter; and then feeding that NBW to the digesters [34]. The NBs can either be on the surface or as free bubbles in the middle of the aqueous phase, in such cases, the bubbles exhibit Brownian motion [35, 36]. The supplementation of NBW needs to be carefully executed in order to balance the antioxidant ability against microorganisms and over-aeration [37]. To date, NBW-supplemented AD studies have been primarily performed in lab-scale batch reactors (60–400 mL) [29, 38], underscoring the need for further research to facilitate its full-scale application.

Recent review articles featured the advantages of incorporating the NBW in AD systems [39–41]. However, these reviews have predominantly emphasized performance evaluation, offering limited analysis of the fundamental mechanisms by which NBW influences AD systems. Understanding these mechanisms necessitates investigating how NBW affects the individual steps of the AD process. Accordingly, the objectives of this review are to provide a comprehensive analysis of recent findings on NBW-integrated AD systems, with an emphasis on the impact of NBW on each step of the AD process and the influence of the operational conditions. Furthermore, this review examines the integration of NBs with other enhancement strategies and evaluates the energy efficiencies of NBW-integrated AD systems. Finally, prospects and the research gaps are discussed. The novelty of this review lies in presenting a cohesive perspective on the benefits, mechanisms, operational considerations, limitations, and future opportunities associated with applying NBW in AD systems.

Impact of Nanobubbles on Anaerobic Digestion

Table 1 summarizes the key findings reported in previous studies that applied NBW in AD systems. As shown in Fig. 1, NBs technology can significantly contribute to enhancing the methane yield, biogas production, and activity of key enzymes in the AD processes. However, these efficiency gains are highly dependent on the type of feedstock, the type of gas used for NB, and the operational conditions of the digesters. It appeared that NB's technology can be effective in degrading various complex substrates, such as swine manure and lignocellulosic biomass (e.g., corn straw) (Fig. 1(a) and 1(b)). Generally, the enhancement in the methane yield varies between 10 and 45% (Fig. 1(a)), while methane increased by 8–44% (Fig. 1(b)). Moreover, the enzymes activity can be significantly enhanced by introducing NBs technology in AD, where the activity of enzymes and coenzyme F_{420} can be enhanced by approximately 60% based on the gas type and wastewater source (Fig. 1(c)).

Applying NBW to AD can significantly enhance process efficiency through various mechanisms. The high zeta potential, low buoyancy, and small size (< 1 µm) of NBs can improve water mobility, allowing them to effectively transport essential nutrients to microbes, promote their growth and enzyme activity, and enhance the mass transfer efficiency for gaseous species to microbial flocs [55, 66]. Notably, O_2-NBW can stimulate higher adenosine triphosphate (ATP) synthesis and metabolite transport processes [34]. Carbon dioxide (CO_2) and H_2-containing NBs can act as additional substrates for methanogens [40], enhancing the overall performance of AD systems. The following two sections will provide more insights on the impacts of NBs into the individual steps of AD.

Impact of Nanobubbles on Hydrolysis and Acidogenesis

NBW can substantially improve the hydrolysis step in digesters, which is considered the rate-limiting step of AD in complex feedstock [39, 41, 67]. Therefore, it has been applied to enhance the biodegradation and hydrolysis reactions of complex substrates, such as lignocellulosic biomass and food waste [60]. Supplementing air-NBs has been shown to improve the hydrolysis rate due to limited micro-aerobic conditions [55, 58]. The hydrolysis-implicated key enzymes could be enhanced by NBW, leading to the acceleration of the hydrolysis of refractory compounds [56]. The NB-induced dissolved oxygen was found to act as a reagent in the first step of the aromatic ring opening, a process in which the stable benzene-like structures are cleaved by oxidative reactions [68], and boost the diversity of facultative hydrolytic bacteria [28] (Fig. 2). Moreover, NBW can increase the activity of hydrolytic extracellular enzymes and the facultative bacterial diversity, enhancing the performance of acidogenesis process. In addition, the radicals generated by NBs (e.g., •OH and •H radicals) can increase the oxidative decomposition of substrates by the degrading microorganisms [60] (Fig. 2). Importantly, such a positive effect of free radicals is related to their limited production, as excessive radical production can lead to cell wall disruption and DNA destruction [69]. Given that the accumulation of intermediate compounds (VFAs) inhibits the AD process [70], the consumption of VFAs was also found to be accelerated by supplementing NBW, thereby stabilizing the AD system by alleviating the risk of acidification. It was previously presumed that NBs can improve ATP generation, whereas it has been experimentally proven that NBs enhance the activity of the electron transport system (ETS) [34, 41]. ETS occurs in the bacterial cell membrane and is responsible for electron transfer in oxidative phosphorylation, which is a metabolic electron transport pathway that generates ATP. Particularly, electrons are transferred from coenzymes, generated by the Szent-Györgyi-Krebs cycle, such as nicotinamide adenine dinucleotide (NAD^+) and nicotinamide adenine dinucleotide-hydrogen (NADH), to the terminal electron acceptors [71, 72]. NBs have a negative surface charge in water, so they attract ions with the opposite charge, thereby achieving neutralization [73]. Expectedly, NAD^+, along with protons (H^+) are attracted to NB. Considering the high speed of motion of NB, faster NAD^+ movement could be acquired, leading to a higher generation of NADH (the reduced form of NAD^+) and ATP [34] (Fig. 2). NAD(H/+) is a key cofactor in the AD process. NADH establishes the reductive environment for intracellular reductions, whereas NAD^+ acts as an electron acceptor, resulting in the regeneration of NADH. The NADH oxidation process releases a large amount of energy, causing a proton gradient that allows for the generation of ATP and the exchange of metabolites [74]. Another potential reason for the NB-induced increase in ATP, particularly in the case of O_2-NB, could be the collision of NBs resulting in collapse. This collapse leads to the release of O_2; thus, the generated micro-O_2 environment might promote heightened ATP synthesis [34] (Fig. 2).

In addition to their impacts on hydrolysis, it was found that NBs can promote acidogenesis and acetogenesis steps in AD. Specifically, NBs can weaken the hydrogen bond network in bulk water by absorbing water protons. This results in a reduction in network integrity, a decrease in water viscosity, and an increase in the mobility of water molecules [75]. Then, an acceleration in the mass transfer

Table 1 Impact of nanobubbles water supplementation on anaerobic digestion

Tested parameter	Substrate	Gas type	NBW Dosage[1]	Impact (%, compared to DW)	Reference
Impact of NBW on AD	Corn straw	$CH_4 + CO_2$	~1:3	≤17% ↑ (CH_4 production) 7% ↑ (β-glucosidase activity) 11% ↑ (F_{420} activity)	[42]
		CO_2	NA (Final TS of 5%)	7% ↑ (CH_4 production)	[43]
		O_2		18% ↑ (CH_4 production)	
		CO_2	5:8	4%–17% ↑ (CH_4 production) 23% ↑ (Electron transport system activity) 15% ↑ (β-glucosidase activity) 23% ↑ (Dehydrogenase activity) 11% ↑ (F_{420} activity)	[44]
		H_2	5:8	12%–25% ↑ (CH_4 production) 4% ↑ (CH_4 content) 20% – 33% ↑ (Cellulose reduction) 13% – 26% (Hemicellulose reduction) 9% – 21% ↑ (VS removal) 7% – 28% ↑ (TS removal)	[45]
		N_2	1:2	10% ↑ (CH_4 production) 13% ↑ (Cellulose reduction) 6% ↑ (Hemicellulose reduction)	[30]
		O_2	1:2	22% ↑ (CH_4 production) 18% ↑ (Cellulose reduction) 6% ↑ (Hemicellulose reduction)	
Impact of NBW on co-digestion	Kitchen waste + sewage sludge	Air	~1:3	5% ↑ (CH_4 production)	[46]
		CO_2		25% ↑ (CH_4 production)	
		$Air + CO_2$		21% ↑ (CH_4 production)	
	Cow manure + cabbage waste + PLA	Air	NA (Final TS of 10%)	13–15% (CH_4 production)	[47]
Impact of NBW on AD	Dewatered sludge	Air	~1:2	23% – 42% (CH_4 production)	[48]
		Air	1:2.2	28% ↑ (CH_4 production)	[49]
		CO_2		32% ↑ (CH_4 production)	
		He		16% ↑ (CH_4 production)	
		N_2		9% ↑ (CH_4 production)	
Impact of two different NBW (N_2 and air) on two-stage AD	Food waste	N_2	1:2.1	24% ↑ (H_2 production) 7% ↑ (Glucose reduction) (day 12) 23% ↑ (α-glucosidase activity) 12% ↑ (Protein degradation) (day 8)	[50]
			1:7.5	16% ↑ (CH_4 production) 4% ↑ (F_{420} activity) 4% ↑ (VS reduction)	
		Air	1:2.1	40% ↑ (H_2 production) 14% ↑ (Glucose reduction) (day 12) 14% ↑ (α-glucosidase activity) 16% ↑ (Protein degradation) (day 8)	
			1:7.5	24% ↑ (CH_4 production) 34% ↑ (F_{420} activity) 8% ↑ (VS reduction)	

Table 1 (continued)

Tested parameter	Substrate	Gas type	NBW Dosage[1]	Impact (%, compared to DW)	Reference
Using NBW (N_2, O_2 and CO_2) for soaking the rice straw	Rice straw	N_2	10:11	79% ↑ (VFAs production)	[51]
		O_2		102% ↑ (VFAs production)	
		CO_2		214% ↑ (VFAs production)	
Using NBW (N_2, O_2 and CO_2) as AD accelerator	Rice straw	N_2		0% ↑ (CH_4 production)	
		O_2		6% ↑ (CH_4 production)	
		CO_2		7% ↑ (CH_4 production)	
Use of sludge augmented with NBW (CO_2 and N_2) for high solid AD	Corn straw	O_2	1:2	47% ↑ (CH_4 production) ~4% ↑ (Cellulase activity) ~54% ↑ (Xylanase activity) ~18% ↑ (Ligninolytic activity)	[38]
		N_2	1:2	18% ↑ (CH_4 production) ~3% ↑ (Cellulase activity) ~26% ↑ (Xylanase activity) ~77% ↑ (Ligninolytic activity)	
Impact of NBW on AD stability	Swine manure	Air	1:1	12% ↑ (CH_4 production) 3% – 15% ↑ (Extracellular hydrolytic enzymes activity) 9% – 21% ↑ (F_{420} activity)	[52]
	Cattle manure	Air	1:2.2	10% ↑ (CH_4 production) (Mesophilic condition)	[31]
				18% ↑ (CH_4 production) (Thermophilic condition)	
Impact of NBW on AD stability and CH_4 production	Pig manure	Air	1:2	12% – 17% ↑ (CH_4 production) 4% – 5% ↑ (VS reduction) NBW improved the production/consumption of VFAs even under acidic inhibition	[53]
Impact of CO_2-NBW on high solid AD	Pig manure	CO_2	1:2	20% – 25% ↑ (CH_4 production)	[54]
Effect of three different NBW (N_2, O_2 and air) on AD	Cellulose	N_2	1:2	9% ↑ (CH_4 production) 10% ↑ (Cellulose reduction) 6% ↑ (Cellulose crystallinity reduction) 19% ↑ (Cellulase enzyme activity)	[34]
		O_2	1:2	30% ↑ (CH_4 production) 12% ↑ (Cellulose reduction) 7% ↑ (Cellulose crystallinity reduction) 19% ↑ (Cellulase enzyme activity)	
		Air	1:2	8% ↑ (CH_4 production) 7% ↑ (Cellulose reduction) 4% ↑ (Cellulose crystallinity reduction) 38% ↑ (Cellulase enzyme activity)	
		N_2	19:20	21% ↑ (CH_4 production)	

Table 1 (continued)

Tested parameter	Substrate	Gas type	NBW Dosage[1]	Impact (%, compared to DW)	Reference
Effect of three different NBW (air and CO_2) on AD	Cellulose	Air	1:2	18% ↑ (CH_4 production) 30% ↑ (VS reduction) 29% ↑ (Cellulose reduction) 78% ↑ (Cellulose crystallinity reduction)	[55]
		CO_2	1:2	10% ↑ (CH_4 production) 20% ↑ (VS reduction) 21% ↑ (Cellulose reduction) 56% ↑ (Cellulose crystallinity reduction)	
Impact of N_2-NBW on AD	WAS	N_2	1:2	29% ↑ (CH_4 production) 14% – 17% ↑ (Hydrolases activity) 0.5% ↑ (VS reduction) 48% ↑ (Proteins reduction) 262% ↑ (Carbohydrates reduction)	[56]
Impact of three different NBW (CO_2, air and H_2) on AD	Thermal-hydrolysis pre-treated sewage sludge	CO_2	1:17	37% ↑ (CH_4 production)	[57]
		Air		26% ↑ (CH_4 production)	
		H_2		15% ↑ (CH_4 production)	
Effect of three different NBW (air, N_2, H_2 and CO_2) on AD	WAS	Air	1:2	18% ↑ (CH_4 production)	[58]
		N_2		14% ↑ (CH_4 production)	
		H_2		21% ↑ (CH_4 production)	
		CO_2		20% ↑ (CH_4 production)	
Effect of two different NBW (N_2 and CO_2) on co-digestion	Acetic acid + Lignin	N_2	NA	22% ↑ (CH_4 production) 13% ↑ (Lignin degradation)	[59]
		CO_2	NA	17% ↑ (CH_4 production) 11% ↑ (Lignin degradation)	
Impact of N_2-NBW on the degradation of lignin, co-digested with WAS	WAS + Alkaline lignin	N_2	1:2	21% – 31% ↑ (CH_4 production) 10% – 16% ↑ (Lignin degradation)	[60]
Solid organics solubilization of and methane production by AD under NBW (air, H_2 and CO_2) addition	Swine manure	Air	1:2	19% ↑ (CH_4 production) ~14% ↑ (Protein degradation) ~5% ↑ (Electric conductivity)	[61]
		H_2	1:2	39% ↑ (CH_4 production) ~14% ↑ (Protein degradation) ~10% ↑ (Electric conductivity)	
		CO_2	1:2	21% ↑ (CH_4 production) ~18% ↑ (Protein degradation) ~4% ↑ (Electric conductivity)	
Impact of NBW on the hydrolytic acidification of AD	Rice straw	CO_2	ND	8% ↑ (CH_4 production) 5% ↑ (Lignocellulosic material degradation) 1% ↑ (VS reduction)	[62]
		N_2		7% ↑ (CH_4 production) 4% ↑ (Lignocellulosic material degradation) 2% ↑ (VS reduction)	
		H_2		4% ↑ (CH_4 production) 3% ↑ (Lignocellulosic material degradation) 1% ↓ (VS reduction)	

Table 1 (continued)

Tested parameter	Substrate	Gas type	NBW Dosage[1]	Impact (%, compared to DW)	Reference
Air-NBW mitigating impact on the inhibition of high salinity (up to 30 g NaCl/L), while the co-production of hydrogen and methane from two-stage AD	Food waste	Air	1:2	14% – 43% ↑ (CH_4 production) 21% – 65% ↑ (H_2 production) 0.4% – 22% ↑ (Protease activity) 9% – 20% ↑ (α-glucosidease activity) 26% – 58% ↑ (F_{420} activity)	[63]
Impact of KOH supplementation on salt (20 g NaCl/L) tolerance of methanogenesis in a two-stage AD using pre-acclimated sludge by NBW	Food waste	Air	2:5	17%–25% ↑ (CH_4 production) 46% ↑ (H_2 production) 27% ↑ (Inoculum activity)	[64]
Impact of thermal pretreatment and air-NBW on AD	Cowpea hull + WAS	Air	19:20	Negligible impact on CH_4 production	[29]
		N_2	19:20	21% ↑ (CH_4 production)	
Impact of CO_2-NBW and iron nanoparticles (0.5 g/L) on AD	Corn straw	CO_2	5:8	26% ↑ (CH_4 production) 15%–20% ↑ (F_{420} activity) 20%–28% ↑ (F_{420} activity) 20%–28% ↑ (Electron transport system activity) 10% ↓ (Reducing sugar concentration) 26% ↑ (TS reduction) 17% ↑ (VS reduction)	[65]

[1]NBW volume: working volume

• NBW, nanobubble water; WAS, waste activated sludge; VS, volatile solids; TS, total solids; TOC, total organic carbon; VFAs, volatile fatty acids; AD, anaerobic digestion; ND, not determined.

• All studies were implemented using lab-scale batch AD reactors.

of nutrients from the aqueous solution to the microbial cells can be achieved. Consequently, the production of VFAs is improved [58] (Fig. 3).

By combining O_2-NBW supplementation and bioaugmentation technologies, the enzyme activities of cellulase, lignin peroxidase, and xylanases were improved by 2%–55%, while the methane production rate increased by 47%[38]. Notably, bioaugmentation is a recovery approach used to enhance the AD system's performance after exposure to high organic loading rates or toxicity. Bioaugmentation is done by adding specific microbes to the digester system [76]. In another study, the methane yield was enhanced by 10% and 22%, compared to control reactors, when corn straw was pre-augmented using N_2-NBW and O_2-NBW reactors, respectively [30]. Moreover, the biodegradation rate was 20%, 33%, and 38% for cellulose and 29%, 35%, and 35% for hemicellulose using the control, N_2-NBW, and O_2-NBW reactors, respectively.

Impact of Nanobubbles on Methanogenesis

NBW can contribute to the establishment of direct interspecies electron transfer (DIET), which enhances the overall performance of anaerobic digesters (Fig. 4). The release of O2 from O2-NBW acts as a vigorous electron acceptor, thereby enhancing microbial activities and expediting electron transfer correspondingly [34]. On the other hand, multiple studies have reported that the NBW technology can positively affect the growth of methanogens when air, O_2 or CO_2 is supplemented in NBW, significantly enhancing the methanogenesis step in the AD processes [50, 55]. In contrast, a few reports suggested a negligible influence of N_2-containing NBs on methanogenesis [50, 56]. However, the efficiency improvements from applying NBW technology in AD can vary significantly depending on the waste sources, reactor types, and the NBW dosage used (Table 1). For example, it was found that the addition of N_2-NBW enhanced the methane yield by 29% compared to the control samples during the AD of waste activated sludge (WAS) [56, 72]. In addition, Yang et al. [56] reported that N_2-NBW increased the activity of four extracellular hydrolases (e.g., alkaline and acid phosphatase) by 14–17% during AD. Moreover, it was found that the N_2-NBW increased the negative charge of sludge particles compared to the deionized water (DW) control. Then, the augmentation of the sludge hydrolysis process was claimed as the main mechanism for enhancing the methane production during the AD process [56]. In addition, N_2-NBW enhanced the total methane yield

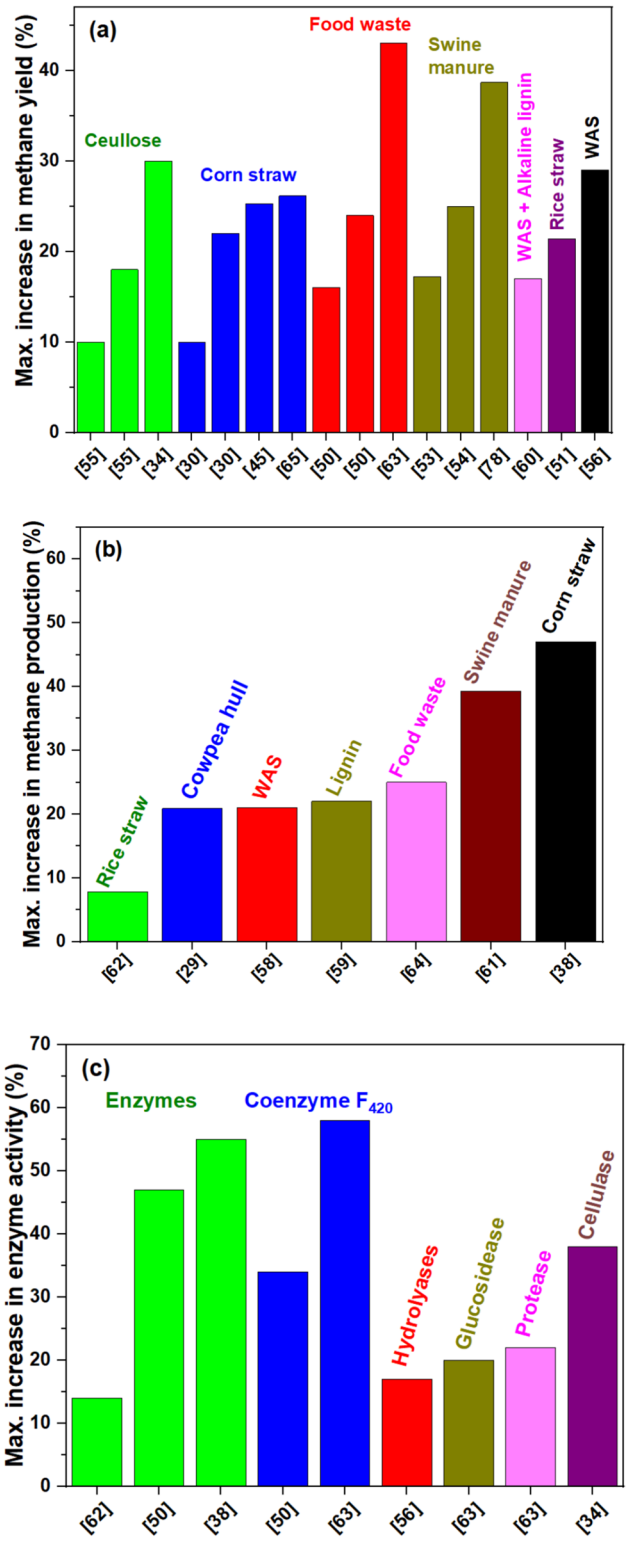


Fig. 1 Efficiency gains from incorporating nanobubble technology to enhance: **a** methane yield; **b** methane production; and **c** enzyme activity; WAS stands for waste activated sludge

by 17% when supplemented during the anaerobic co-digestion of WAS and alkaline lignin [60]. The lignin degradation rate was considerably increased, with a noticeable enhancement in volatile solids (VS) reduction, when N_2-NBW was added to batch AD experiments [60].

In multiple previous lab-scale studies, the cumulative methane yield was enhanced by 12.1–38.7% compared to the control due to the supplementation of air- and CO_2-NBW for AD of ammonia-rich manure [53, 54, 61]. Moreover, NBW could significantly improve the microbial diversity and abundance of methanogens (e.g., *Methanosaeta* and *Methanospirillaceae)* [51, 65], enhancing the methanogenic activity. Notably, air-NBW enhanced both hydrolysis/acidification and methanogenesis reactions in the AD of food waste. Moreover, the energy recovery was significantly increased by 23% [50]. In another study, air-NBW was combined with a two-stage AD of food waste to minimize the effects of salt (NaCl) inhibition on the production of hydrogen in the hydrolysis/acidification and methane in the methanogenesis stages during the AD of food waste [63]. It was demonstrated that the air-NBW addition enhanced the hydrogen yield by 21–65% while the methane yield was increased by 14–43% compared to the DW control reactors. The activity of protease and α-glucosidase was also increased by 0.4–22% and 9–20%, respectively, in the first reactor. In addition, the activity of coenzyme F_{420} was elevated by 26–58% in the methanogenesis reactor, by the addition of air-NBW. F_{420}, a coenzyme found in methanogens, plays a crucial role in the methane production process. Consequently, the activity of F420 is widely used to reveal important information about methanogens, such as their growth rate and morphological type [77]. In another study, air-NBW combined with an alkaline agent (KOH) was used in a two-stage AD system of food waste where hydrogen yield was improved by 46%, while methane production increased by up to 25% [64]. Moreover, NBW improved the activity of dehydrogenase and coenzyme F_{420}, compared to the control reactors, due to the elevated absolute value of the zeta potential and spin–spin relaxation time, which are used to characterize the diffusion and mobility of water molecules in biological fields [61]. Overall, NBs have positively impacted coenzyme activities and the abundance and diversity of methanogens (Fig. 4). Furthermore, nanobubbles may enhance the release of dissolved CH_4, thereby increasing CH_4 recovery, which warrants further investigation to better understand the underlying mechanisms.

In general, supplementing NBW can boost microbial activity [78]. However, multiple studies have referred to a specific feature related to O_2, CO_2, and H_2 NBs, as they can enhance the relative abundance of hydrogenotrophic methanogenesis, thereby improving the AD process performance [31, 42, 79]. Specifically, CO_2 and H_2 NBs can be directly utilized as substrates by

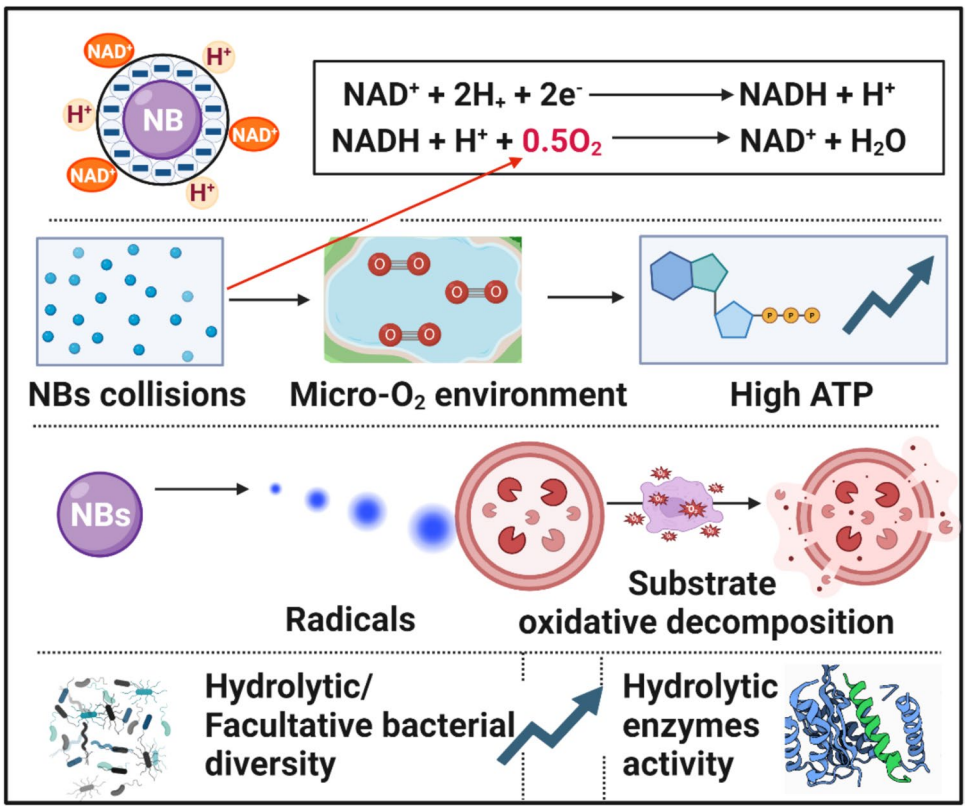


Fig. 2 Schematic diagram showing the impact of nanobubbles (NBs) on hydrolysis (Created with Biorender)

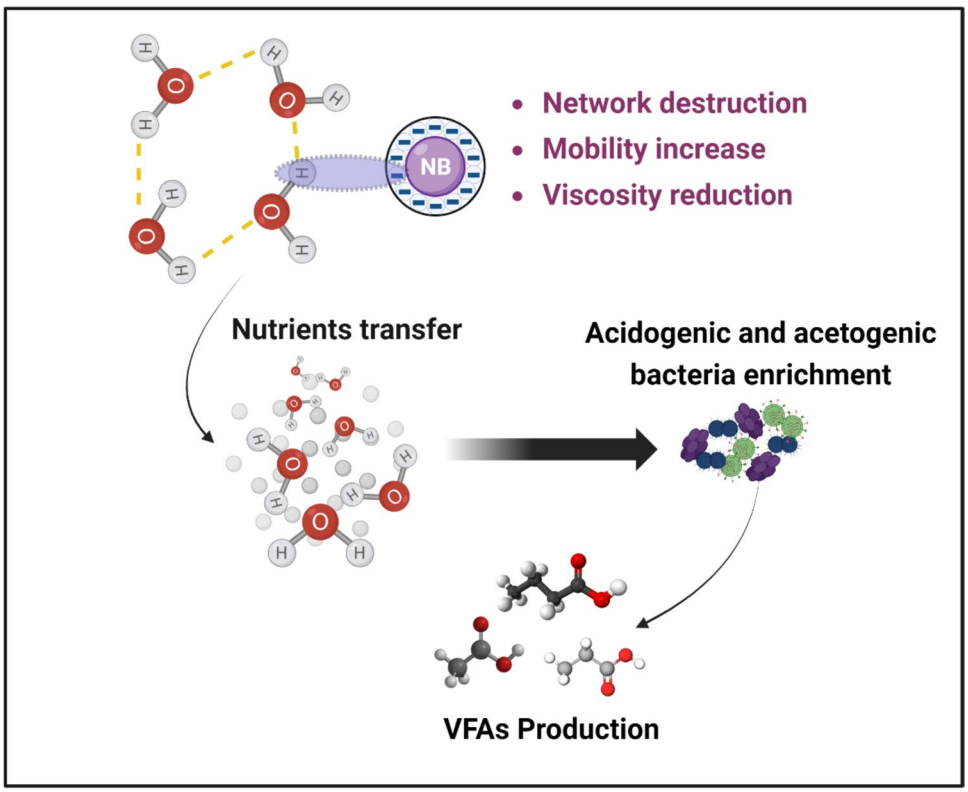


Fig. 3 Schematic diagram showing the impact of nanobubbles (NBs) on acidogenesis/acetogenesis (Created with Biorender)

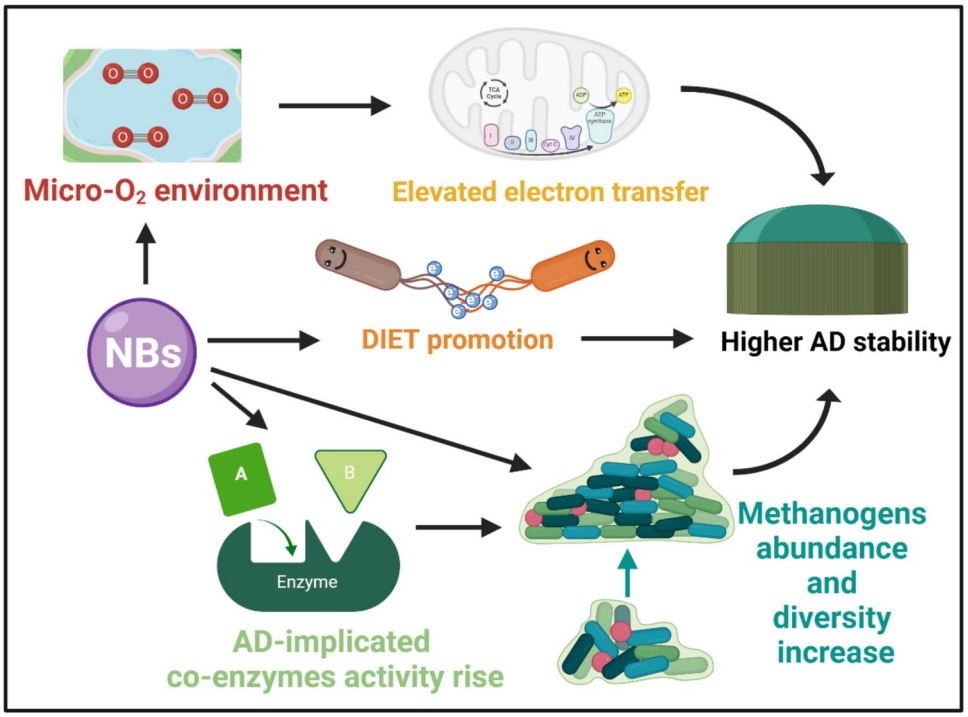


Fig. 4 Schematic diagram showing the impact of nanobubbles (NBs) on methanogenesis (Created with Biorender)

hydrogenotrophic methanogens [40, 80]. Furthermore, since molecular hydrogen can act as an electron donor for hydrogenotrophic methanogens, H_2-NBW could be utilized for biogas upgrading [81]. However, H_2-NBW supplementation can enhance the high hydrogen partial pressure [30]. Such a high hydrogen partial pressure, if not properly controlled, can thermodynamically inhibit obligate hydrogen-producing acetogens, leading to acid accumulation and AD disruption [82]. On the other hand, CO_2 NBs can be readily taken up by the microbes, promoting hydrogenotrophic methanogenesis [31]. Hou et al. [79] reported that the abundance of hydrogenotrophic methanogens in the CO_2-NBW (0.09%) was relatively higher than that of the air-NBW supplemented AD (0.04%), while the abundance of hydrogenotrophic methanogens was zero in the control reactor. Similarly, the supplementation of $(CH_4 + CO_2)$-NBW boosted the relative abundance of *Methanomassiliicoccaceae*, a hydrogenotrophic methanogenic archaea, by 14% [42]. Differently, Pei et al. [28] found an increase in the abundance of hydrogenotrophic *Methanobacterium* in the microaerobic environment created by O_2-NBW supplementation. Such conditions can help achieve syntrophy between facultative bacteria and hydrogenotrophic methanogens [83]. In a recent study, Wang et al. [43] found that CO_2-NBW increased the expression of genes encoding enzymes involved in hydrogenotrophic methanogenesis, while O_2-NBW enhanced both acetoclastic and hydrogenotrophic methanogenesis. The impact of various NBs on microbial metabolic activity and community structure within the AD should be investigated more closely.

Impact of Nanobubbles on Anaerobic Digestion of Lignocellulosic Waste

Due to the positive impact on the hydrolysis step, several studies focused on utilizing NBW in the AD of lignocellulosic wastes (Table 1). NBs were hypothesized to penetrate the oligosaccharide layer of lignocellulosic biomass, which is generated from the incomplete depolymerization of cellulose and hemicellulose [84]. The high surface area provided by NBs can facilitate such penetration. Oligosaccharide layer destruction can ultimately enhance the hydrolysis of lignocellulosic biomass [34]. Furthermore, treatment with NBW can lead to a higher abundance of hexokinases, which are intracellular enzymes responsible for catalyzing the phosphorylation of glucose, fructose, and mannose into hexose 6-phosphates. These resulting phosphate esters can then be metabolized into pyruvate through glycolysis or directly used for biosynthetic processes. Moreover, the supplementation of CO_2-NBW into AD primarily boosted the activities of 1,4-beta-xylosidase and non-reducing end alpha-L-arabinofuranosidase [34, 62]. On the other hand, N_2, O_2, and air NBs can also boost cellulase activity (19%–38% higher than control) [34], although a different study observed minimal impact (< 5% higher than control) [38]. Then, NBW enhances the generation of easily degradable intermediates for the next steps, i.e., acidogenesis, acetogenesis,

and methanogenesis. Further, NBs with high mobility can migrate from the bulk phase to the surface of cellulose particles, thus breaking through the oligosaccharide layer that contributes to mass transfer resistance during the hydrolysis-acidification of lignocellulosic waste. NBs treatments can disrupt the compact structure of rice straw, indicating an enhancement in the biodegradability of rice straw [34, 62]. This disruption explains the reduction in cellulose crystallinity by NBW treatment. A considerable reduction in cellulose crystallinity, measured by crystallinity index, was noticed by supplementing NBW. CO_2- and air-NBW achieved 56%, and 78%, respectively, more crystallinity reduction, compared to the controls. While the achieved NBW-caused increase in cellulose reduction ranged from 7 to 33% (Table 1) [55]. While treating high-solids cattle manure, Zhang et al. [31] denoted that the supplementation of air NBs increased methane yield by 10% and by 18% under mesophilic and thermophilic conditions, respectively, compared to the control. This observation was attributed to the fact that thermophilic conditions can promote faster hydrolysis and acidification, resulting in more efficient degradation of organic matter. The presence of air NBs further amplified this phenomenon by facilitating mass transfer and stimulating microbial activity. A summary of the mechanisms behind biogas yield enhancement by gas type in NB-supplemented AD is presented in Table 2**.**

Nanobubble Water Production and Characterization

Prior to proceeding with the impact of various operational conditions on NBW-supplemented AD systems, it is important to delve into the production of NBW and NBs, as well as the relevant characterization techniques. The generation of NBW and NBs is interrelated as the NBs are generated into homogenous liquid, called NBW in case where water is used [85]. Herein, a brief description of some well-studied generation methods is provided.

Production of Nanobubble Water

Mechanical Methods

A Summary of the major generation methods of nanobubble water NBW is presented in Table 3. For generating NBW, the devices used in mechanical methods consist principally of three components: a water pump, an air compressor, and a nozzle [86, 87]. The two commonly utilized techniques are pressure-based and cavitation-based. In the pressure-based technique, the pump circulates the water that is pulled from the water tank. Bubbles are released at the nozzle, while the air gas is fed to the circulated water from the suction side of the pump. Through the high-pressure system, the water is supersaturated with air. From that water, the NBs are generated due to the pressure gap at the nozzle [86].

On the other hand, the cavitation-based technique depends on the cavitation process. The cavitation process means that water, or any other homogeneous liquid, is subjected to a sudden drop in pressure, leading to the generation of NBs [87]. NBs produced by the cavitation process can be categorized into four classes, depending on the external source used, which can be a probe or a water bath. These classes include optical cavitation (originating from a laser beam), particle-induced cavitation (stimulated by elementary particles), acoustic cavitation (generated by the acoustic field), and hydrodynamic cavitation (caused by ultrasonic waves applied to the flowing fluid) [89, 90]. The considerable parameters of the NBW generation process vary depending on the device used, its configuration, gas type, applied pressure value, gas–liquid ratio, process temperature, and process duration [88, 101]. Recently, the venturi-type NBs generator, a hydrodynamic cavitation-based generator, has garnered significant attention for

Table 2 Summary of biogas/methane yield enhancement mechanisms by gas type in nanobubble-assisted anaerobic digestion

Gas type	Mechanisms behind biogas yield enhancement	Comments	References
Air	• Improves mass transfer efficiency • Enhances hydrolysis process and microbial activity	Moderate improvement in biogas yield	[46, 50]
Oxygen (O_2)	• Enhances activity of hydrolytic and acidogenic bacteria • Increases volatile fatty acids generation	Significant methane yield improvement, especially in high-solids AD; however, requires careful oxygen control	[38, 43]
Hydrogen (H_2)	• Enhances hydrogenotrophic methanogenesis • Promotes direct interspecies electron transfer (DIET)	Must be balanced to avoid acidification; more beneficial in two-stage systems	[45, 63]
Carbon dioxide (CO_2)	• Improves hydrogenotrophic methanogenesis • Facilitates syntrophic interactions	Plays an indirect role in pH stabilization and microbial balance	[62, 65]
Nitrogen (N_2)	• Promotes key hydrolytic enzymes • Enhances hydrolysis/acidification process	Often used as a control gas in experimental studies	[30, 56]

Table 3 Summary of major generation methods of nanobubble water (NBW)

Method Type	Technique	Description	Key features/parameters	References
Mechanical	Pressure-based	Water is circulated under pressure; gas is introduced, and nanobubbles (NBs) are formed at the nozzle due to a pressure drop	Pump, air compressor, nozzle, applied pressure, gas–liquid ratio, temperature, duration	[86–88]
	Cavitation-based	NBs are formed from a sudden pressure drop; includes optical, particle-induced, acoustic, and hydrodynamic cavitation	External source (laser, particles, ultrasound), cavitation device type	[87, 89, 90]
	Venturi-type (Hydrodynamic)	A subtype of cavitation using a Venturi design; commonly used for its simplicity and system compatibility	High reliability, small NB size	[79, 91, 92]
Electrochemical	Electrolysis-based	NBs produced via electrochemical reactions (e.g., water splitting)	Electrode type (e.g., platinum nanodisk), current, voltage, reaction media	[93–98]
Membrane-based	Nano-porous membranes	Gas is injected through nanoporous membranes (50–150 nm pores) submerged in water	Orifice size, gas type, NB size and shape dynamics (departure rate, elongation behavior)	[99, 100]

multiple reasons, including its high reliability, uncomplicated design, reduced maintenance requirements, capability for producing small NBs, and suitability for diverse applications [91, 92]. Due to its simple design, the venturi-type NBs generator can be easily installed into existing AD digesters [79].

Electrochemical and Membrane Methods

In the case of electrochemical methods, NBs are produced through electrochemical reactions occurring at the electrodes. Examples of such reactions include the electrosynthesis of compounds such as sodium chlorate and alkali chlorate [93, 94], the splitting of chemicals in water, the electrosynthesis of organics, and CO_2 electroreduction [95, 96]. The electrolysis of water produces O_2 and H_2 gases; thereafter, the stable NBs of O_2 and H_2 will be formed via nucleation on a platinum nanodisk electrode [97]. NBs of O_2 and H_2 were detected in the commercial electrochemical flow cells, with concentrations and particle sizes ranging from 10^7–10^8 mL^{-1} and 94–529 nm, respectively [98].

On the other hand, the membrane method for generating NBs involves injecting the gas through orifices (with pore sizes between 50 and 150 nm) that are submerged in a water pool [99]. Expectedly, the characteristics of the generated NBs are significantly influenced by the orifices' size. It was noticed that smaller orifices produce NBs that depart faster and tend to maintain the spherical shape. In contrast, larger orifices generate NBs with relatively larger diameters. Those NBs take a longer time to leave the orifices. After exiting from the orifices, generated NBs tend to elongate vertically because of the buoyancy action [100]. Previous review articles comprehensively explain the various NBW generation techniques [37, 101, 102].

Characterization of Nanobubbles in Nanobubble Water

The characterization of NBW quality relies on detecting NBs. NBW contains 10^6–10^8 NBs/mL, while the NBs' diameter is less than 10^3 nm [37]. NB detection is challenging as the NBW is transparent, and it is hard to visualize them using an optical microscope. This issue is more pronounced when dealing with wastewater and sludge, where the presence of complex particles further obscures the detection of NBs, thereby hindering effective process control and monitoring of NBW. Moreover, it is difficult to differentiate between NBs and other nano-sized colloidal dispersions [102]. In this section, the major features used for characterizing NBs and NBW are presented.

Size and Distribution

For determining the concentration of NBs, referring to the size and distribution of NBs, multiple methods can be used, including photographic [103], optical/light scattering [104], acoustic [105], and X-ray particle tracking velocimetry [106]. Among these methods, the photographic method has been most widely used for investigating the size of NBs [86]. Given that it allows for the direct capture and measurement of NBs, the photographic method entails the use of a light source and a high-speed camera. Most of these types of cameras can typically capture more than 500 frames per second [107]. During this capture, the NBs generator is positioned in the center while the camera is positioned opposite to the light source. Automated image analysis uses programming languages such as MATLAB, Octave, Mathematica, and ImageJ [107]. On the other hand, the light scattering technique is another efficient technique for measuring the size and distribution of NBs. The technique employs a photomultiplier tube for identifying the scattered light intensity, which has an inverse relationship with the NBs diameter [108]. Multiple studies that applied NBs in AD systems have adopted the nanoparticle tracking analysis method, a light scattering-based technique for identifying the NBs size and distribution [34, 55, 56, 62].

A previous study investigated the effects of air flow rate and water flow rates on the size distribution of NBs in a Venturi-type generator [109]. The most notable finding was a strong inverse relationship between bubble size and water flow rate. Moreover, increasing the air flow rate boosted the bubble concentration, which in turn promoted the rate of coalescence and resulted in larger bubble size [109]. However, no previous study examined the relationship between gas flow rate and gas type. Reported gas flow rates in the literature varied depending on the gas type, for instance, 0.01 L/min (N_2) [29], 0.2 L/min (H_2) [45], 0.12 L/min (CO_2) [44], and 0.05 L/min (air) [63].

Stability and Zeta Potential

The stability of NBs is evaluated by how long they can remain in the aqueous phase, which ranges from days to months [40]. The stability of NBs can be measured by various tools, including molecular dynamics simulation and zeta potential. Zeta potential refers to the electrical potential at the bubble surface, which affects its interaction with the surrounding environment and overall stability [110]. The zeta potential of NBW is often considered a key indicator of NB stability. This is primarily due to electrostatic repulsion generated by strong surface charges, which helps prevent bubble coalescence and reduces gas dissolution [111]. For instance, NBs exhibit high stability in alkaline conditions because of the formation of a thick electric double layer and a strongly negative zeta potential. Conversely, a less negative zeta potential value, which could be caused by acidic pH, lessens the stability of NBs by weakening surface repulsion forces [112, 113]. Moreover, the presence of salts can further decrease NB stability by shifting the zeta potential toward more positive values [114]. Zeta potential analyzer measures the electrophoretic mobility by detecting the horizontal speed of the NBs. Then, the detected speed is used to calculate the zeta potential via Smoluchowski´s equation. To enhance measurement accuracy, many analyzers involve a video camera and video image analysis software [115, 116]. Examples of widely used zeta potential analyzers include the Zeta Potential Analyzer (Zeecom MicroTech Co. Ltd., Japan), the Zeta-Plus (Brookhaven Instruments Co., New York), the Zetaphoremeter III (Shephy, France) and the ZEN3600 (Malvern Instruments Ltd., UK), among others [117–119].

Numerous theoretical and operational challenges are associated with the stability of NBs. A key theoretical challenge lies in understanding and explaining their unexpectedly high stability. Such stability contradicts the Young–Laplace equation, which expects that high internal pressure of nano-sized bubbles should lead to shrinking and vanishing of the bubbles [111]. On the other hand, multiple experimental studies have consistently shown that NBs can persist for weeks to months [53, 81, 120]. In response to this contradiction, research efforts proposed a mechanism to explain the high stability of NBs. One such proposal suggested that the strong negative charges, on the surface of NBs, prevent their coalescence and create a dynamic equilibrium that hinders gas diffusion, thereby prolonging the bubbles lifetime [121]. Nevertheless, a comprehensive and universally accepted theory to fully explain the stability of NBs is yet to be established. From a practical perspective, another challenge lies in the current NB generation systems, which often lack accurate control over the parameters of the generation process, leading to variations in the stability characteristics of the generated bubbles [111]. The effective integration of NBW technology in AD systems is tied to addressing both the theoretical and practical aspects of NBs'stability, as well as other challenges related to supplementing NBs in AD systems and maintaining consistency.

Impact of Operational Conditions On Nanobubble Water-supplemented Anaerobic Digestion

Impact of Gas Type and Dosages

To date, the NBs technology has been applied with different gases (e.g., air, H_2 and N_2) to evaluate its impact on AD performance [34, 58]. Gas type has implications for the

structure and stability of NBs. The stability of NBs is primarily due to the formation of a double layer around them. This double layer consists of an inner layer of negative charges that are inherent to the NBs, and an outer layer of counter-charges that are attracted to these negative charges. The overall system remains stable due to the neutralization of charges between these two layers [73]. Also, the gas–liquid interface varies based on the NBs'physicochemical properties [68]. The electrical stability of NBs is primarily attributed to their high zeta potential and significant surface charge. Due to their high stability and water solubility, NBs can be maintained in an aqueous solution for several days to months [22]. Zeta potential values, which are indicators for the electrical potential of the NB-liquid interface, are negative in NBW because the hydroxide ions (OH^-) dominate the first molecular layer of the NBW [110, 117]. The zeta potential absolute values were in the range of 20–27 mV, 29–35 mV, 34–45 mV and 17–20 mV for CO_2, N_2, O_2, and air, respectively [117]. The narrow ranges of zeta potential values and the overlapping between these ranges make it impossible to compare the impacts of these gases. The negative value of NBs' zeta potential increases as the pH increases until reaching a threshold of pH 10, beyond which the value decreases [58]. Given that the optimal pH range for the AD process is 6.8–7.4 [122], future research should investigate the implications of NBs-caused pH increases on AD-based wastewater treatment systems.

A previous study demonstrated that the methane yield was enhanced by 21%, 20%, 18%, and 14% using H_2-, CO_2-, air- and N_2-NBW, respectively, compared to control samples using tap water in the AD of WAS [58]. On the other hand, the methane yield was increased by 17% using H_2-NBW, 16% using CO_2-NBW, 14% using air-NBW, and 11% using N_2-NBW compared to control samples using DW in batch reactors for AD of WAS [58]. Importantly, bubble stability followed the descending order as $H_2 > air > N_2 > CO_2$, whereas the negativity of the zeta potential values increased with the increase in pH values (from 3 to 10). Differently, at pH higher than 10, the negativity of the zeta potential values decreased [58]. In another study, multiple types of gases, including air, O_2, N_2, and a mixture of N_2 and O_2, were used for AD of cellulose, where the methane yield was enhanced by 8–30% compared to the control samples using DW [34]. In addition, the addition of NBs increased cellulose content removal by 8–14% and cellulose crystallinity by 9–21%, compared to the control batch reactors [34]. The cellulase activity was also increased by 10–38% due to the supplementation of gas-based NBs, where the O_2-NBW increased the activity of ETS by 1.7 times compared to control samples [34].

When comparing the effect of air-NBW and CO_2-NBW addition on the degradation of cellulose, it was found that the methane yield was elevated by 18% using air-NBW and 10% using CO_2-NBW compared to the control samples in batch AD experiments [55]. In addition, cellulose crystallinity decreased by 14–20%, whereas the microbial activity of hydrolytic bacteria and methanogens increased significantly due to the supplementation of NBW. In another study, air-, H_2-, and CO_2-NBW were added to enhance the AD of swine manure using batch AD reactors, where it was found that these NBs enhanced the biogas production rate by 16–32% and methane production by 19–39% [61]. H_2-NBW provided the highest methane production. Additionally, the improvements in the hydrolysis and acidification reactions enabled NBW to achieve a higher production of VFAs. Air-NBW is expected to support both VFAs production and hydrolysis, which in turn boosts methane production. The utilization of oxygen NBs, produced from a pressure swing adsorption generator, seems to be as a cost-effective alternative to using either air or pure oxygen [68, 123].

The NBW dosage varied among the studies. Most studies used an NBW volume-to-working volume ratio of 1:2 (see Table 1), which appeared to be the optimal ratio providing the most promising results. It is still important to test the substrate-specific effectiveness of various NBW volume-to-working volume ratios to confirm whether they have the potential to enhance the AD process. Nonetheless, at higher ratios, the volume of tested substrate would be limited compared to NBW volume, which is not attractive from an operational perspective. A higher volume of NBW would occupy the working volume of the digester, reducing its overall capacity. Moreover, the process water requirement for NBW generation would be higher.

Some previous studies highlighted that the concentrations of NBs in the NBW are affected by the operating time of the NB generator [58, 124]. In a later study, the impact of different operating times (5, 10, 15, and 20 min) on the NBs concentration and distribution was investigated [51]. The experiment was conducted in a 20 L water tank using an NB generator set at a gas flow rate of 1 L/min. Three gases were tested, O_2, N_2 and CO_2. For N_2-NBW and O_2-NBW, the most appropriate operating time was 15 min, at which the concentration of NBs was set at $1.43 \times 10^8 \pm 1.5 \times 10^7$ particles/mL and $3.8 \times 10^8 \pm 3.7 \times 10^7$, respectively. After 15 min, the NB concentrations were unstable, and the water temperature increased. Comparatively, only 10 min was enough to achieve the highest NBs concentration ($1.1 \times 10^8 \pm 1.0 \times 10^7$ particles/mL) in CO_2-NBW. By exceeding the 10 min operation time, the NBs concentration dropped. This variance among CO_2, O_2, and N_2 can be explained by the nature of the gas. For instance, CO_2 is more soluble than O_2 and N_2, which may lead to faster dissolution in water in the case of CO_2 [125, 126]. This can justify the shorter optimal time and lower concentration in the case of CO_2, compared to O_2 and N_2. Another potential reason is the formation of carbonic acid through the reaction of water with CO_2, which

consumes a portion of the fed CO_2 [127]. Further research studies would be required to investigate this aspect.

Impact of Oxidation–reduction Potential

Oxidation–reduction potential (ORP) is typically measured in anaerobic systems to indicate electron transfer and redox balance [128]. Moreover, the key indicators used for monitoring the low concentrations of dissolved oxygen (DO) in the microbubbles-supplemented anaerobic digester are the ORP and DO [68]. Expectedly, the measurement of DO is more accurate than ORP; however, obtaining accurate and stable values for DO is very challenging, especially when considering the unequal distribution of DO in the case of improper operation of a continuously stirred tank reactor [129, 130]. Another problem in measuring DO is the high likelihood of rapid consumption of dissolved oxygen by facultative bacteria [83]. Additionally, the standard DO meter is not efficient for measuring low DO concentrations in NBW-supplemented anaerobic digesters [131]. On the other hand, ORP measurement secures an easy and precise indicator for DO amount, in light of the logarithmic relationship between DO and ORP, whereas the ORP value of − 50 mV refers to the DO of 0.1 mg/L [10]. Using ORP meters, any slight variance in oxygen concentration in the liquid solution can be detected [15]. Furthermore, targeting the ORP value allows for controlling the digestion. For instance, a previous study indicated that targeting an ORP of −230 mV effectively removes both gaseous and dissolved hydrogen sulfide, while also increasing methane yield by 56.3% compared to the control [132].

Low ORP values generally refer to the stricter anaerobic conditions [133]. Typically, the range of ORP values in an anaerobic condition is 200–400 mV, by a standard hydrogen reference electrode [134]. Facultative microorganisms begin to reproduce when the oxidation–reduction potential (ORP) rises above −200 mV, while aerobic microorganisms proliferate once the ORP exceeds 0 mV. On the other hand, at the ORP range (0 to −300 mV), the facultative microorganisms produce intermediate products, such as VFAs, which are later consumed by methanogens [15]. Methanogens, which are strict anaerobes, thrive at ORP lower than −400 mV [83]. Microaerobic conditions, achieved by supplementing micro-sized gas bubbles, can reduce the ORP to approximately −470 mV. This enhances both VFA production and methane productivity [10, 15]. While treating 2-butenal manufacturing wastewater, microaeration lowered the ORP to −315 mV, ensuring advanced and steady performance for an expanded granular sludge bed reactor [135]. Another study [34] reported that the supplementation of deionized water (DW) to the cellulose treating AD system led to achieving an ORP value of -489 ± 1.4 mV. Such ORP value increased to -480 ± 5.7 mV, -475 ± 1.4 mV and -412 ± 14 mV, while using N_2-, air- and O_2-NBW, respectively. Furthermore, applying NBW yielded methane yields 8–30% higher than those of DW-based controls. The highest methane enhancement (30%) was acquired using O_2-NBW [34]. Under anaerobic conditions, the pure O_2-NBW had a minimal ORP value of −412 mV [34]. However, if the supplementation of O_2-NBW is excessive, available oxygen can exceed the consumption needs of facultative bacteria. Then, high concentrations of ROS can be formed, leading to damage to the microbial phospholipids and nucleic acids, which causes a decrease in methane production [133]. For AD of corn straw, another study found that the ORP of H_2-NBW reactor was less than that in the control reactor [45]. Furthermore, H_2-NBW was found to enhance methane production by increasing the consumption of reducing substances by methanogenic consortia.

Microaerobic conditions were previously reported to double the abundance of the facultative bacterium *Proteiniphilum* spp., improving the conversion of VFAs to CO_2 in the AD process [10]. Moreover, limited microaeration (DO of 0.2–0.3 mg/L, ORP of −210 mV) could provide H_2S removal [136]. Despite this, adopting ORP-based control of the NBW-supplemented AD process has several challenges [137]. ORP values fluctuate as a result of any changes in operational parameters, such as temperature, pH, toxic compounds and ionic strength. The increase in toxic organics (e.g., chloroform), in the anaerobic environment leads to an increase in the noticed ORP measurement as well [135, 138]. For instance, the rise in toxicity was found to cause a dramatic increase in ORP value from −269 mV to −79 mV[135]. Therefore, achieving stable operation for the NBW-supplemented AD requires the adoption of accurate monitoring and control systems that automatically respond to changes in operational conditions. Notably, ORP measurement can be affected by external electric, magnetic, or electromagnetic fields [139]; further research is needed to mitigate such interference. Integrating machine learning (ML) and artificial intelligence (AI) can help predict process fluctuations, optimize parameters, and provide adaptive control strategies to achieve and maintain optimal performance in AD systems. Incorporating data from the ORP sensor, along with measurements from real-time monitoring of other parameters, such as pH and electrical conductivity, into an ML-based model can enhance the accuracy of predictions for almost all AD process parameters [140]. The deep learning model used in this study combined two models: bidirectional long short-term memory and a convolutional neural network [140]. Such deep learning models can be applied for AD integrated with NBs.

Nanobubbles Integration With Other Technologies

Integrating diverse AD process intensification technologies has proven successful in previous studies [141]. In the same regard, NBs technology has been integrated with other AD process intensification technologies (Fig. 5). For example, air-NBW and N_2-NBW were integrated with two-stage AD to maximize the degradation of food waste and the production of hydrogen and methane [50]. In the first stage (hydrolysis/acidification for hydrogen production), hydrogen yield was increased by 38%, while the methane yield was elevated by 24% using air-NBW and 16% using N_2-NBW compared to the control samples with DW in the second stage (methanogenesis for methane production). In addition, air-NBW enhanced the activity of four extracellular hydrolases (e.g. acid and alkaline phosphatase and protease) by 8–47% in the first stage, while it elevated the activity of coenzyme F_{420} by 34% in the second stage [50]. In another study, the anaerobic sludge was acclimated for 30 days prior to the AD process [38]. During the bioaugmentation, half of the working volume of the utilized batch bottle was filled with sludge, while the other half was filled with either O_2-NBW or N_2-NBW, with corn straw used as the sole substrate. After the bioaugmentation period is over, the resultant sludge is centrifuged, and a certain weight of the resultant sludge is used as seed in a high-solids AD batch test. It was found that the activity of hydrolytic/acidogenic bacteria and methanogenic archaea was more enhanced with the supplementation of O_2-NBW compared to N_2-NBW [38]. NBW-bioaugmented sludge can enhance the degradation of cellulose and lignin, accompanied by an increased abundance of hydrolytic/acidogenic bacteria and methanogenic archaea [38]. However, a few previous studies [50, 56] have highlighted that the N_2-NBW can only enhance hydrolysis with a negligible influence on methanogenesis. However, the variance in the results can be attributed to the differences in feedstocks (corn straw vs. WAS) and other AD operating parameters.

The NBW technology can also be integrated with iron nanoparticles to enhance methane production and increase methane content in biogas. In a previous study, these two approaches were combined to increase the methanogenesis kinetics during the AD of corn straw using CO_2-NBW [65]. The methane yield was enhanced by 26% compared to the control reactors, where enzyme activity and ETS significantly improved. In addition, combining CO_2-NBW and iron nanoparticles enhanced the microbial diversity and abundance of methanogens in the digester. Moreover, the N_2-NBW technology was combined with thermal pre-treatment for improved co-digestion of cowpea hull and WAS, resulting in a 20.9% enhancement in methane production [29].

Energy Efficiency

Calculating the energy consumption and economic feasibility of supplementing NBW into AD systems is essential for evaluating its applicability in large-scale digesters. Unfortunately, such calculations were performed in a very

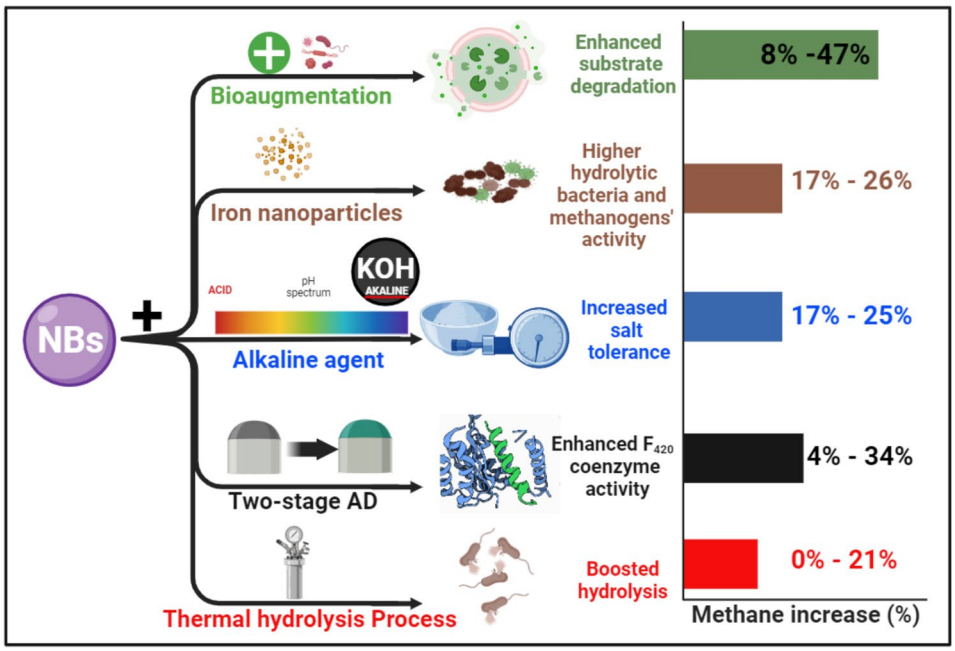


Fig. 5 Integration between nanobubbles (NB) with other anaerobic digestion intensification technologies. The range of methane production increases for different conditions was adopted from the literature [29, 52, 65] (Created with Biorender)

limited number of studies. Hou et al. [50] calculated the energy recovery obtained from the separate production of hydrogen and methane from a two-stage AD of food waste, using Eq. (1).

$$E_r = 12.78 \times Y_{Hydrogenyield} + 40.03 \times Y_{Methaneyield} \quad (1)$$

where E_r, $Y_{Hydrogen\ yield}$ and $Y_{Methane\ yield}$ refer to energy recovered from the reactor, hydrogen yield harvested from the first stage and methane yield harvested from the second stage, respectively. While the 12.78 kJ/L and 40.03 kJ/L are the heating values for hydrogen and methane, respectively, at a temperature of 25 °C and pressure of 1 atm [142]. Hou et al. [50] reported that the energy consumed for operating an NBs generator at a production capacity of 12–22 L/min was 0.75 kW. Assuming the treatment of 1 ton of food waste, it was found that 3.31 kWh of electricity was required to produce 4500 L of air-NBW required for AD operation. Two scenarios, AD of food waste along with DW or air-NBW, were compared. AD was performed in two steps: hydrogen and methane production. Air-NBW scenario revealed 38%, 24%, and 23% higher hydrogen yield, methane yield, and electricity generation, respectively, compared to DW-fed AD [50].

Net energy gain was calculated in a recent study that investigated the impact of utilizing O_2- and N_2-NBW bioaugmented sludge in a high-solids AD of corn straw [38]. The calculation was established based on assuming the treatment of 1 ton VS of corn straw. Specifically, the digester input energy (E_{input}) and digester output energy (E_{output}) were calculated based on Eq. (2) and (3), respectively [143]. On the other hand, the net gain energy (ΔE) and energy recovery (η) were calculated based on Eq. (4) and (5), respectively [143].

$$E_{input} = mC\left(T_d - T_a\right) + 5\%mC\left(T_d - T_a\right) \quad (2)$$

$$E_{output} = q \times P_{methane} \times \zeta \quad (3)$$

$$\Delta E = E_{output} - E_{input} \quad (4)$$

$$\eta(\%) = \frac{\Delta E}{E} \times 100\% \quad (5)$$

where E_{input}, m, and C refer to digester's input energy, digestate mass, and sludge specific heat capacity (4.18 MJ/t/°C), respectively. T_d and T_a are the digestion temperature (applied as 35 °C) and the ambient temperature (considered as 25 °C), respectively. A 5% value was applied in the equation to mimic the heat loss from the high solids anaerobic digester during AD [144]. E_{output}, q, $P_{methane}$ and ζ refer to output energy recovered from AD of corn straw, energy conversion efficiency (assumed as 90%), methane yield from the digester and methane lower heating value (35.8 MJ/m^3), respectively. ΔE stands for the net energy gain or the energy balance, calculated as the difference between the output and input of energy. E term used in Eq. (5) refers to the corn straw heating value, in case of its direct combustion. The highest net energy gain of 5138 MJ/t VS, which was equivalent to an energy recovery of 34%, could be achieved by the reactor with O_2-NBW bioaugmented sludge [38]. For N_2-NBW and DW bioaugmented sludge containing reactors, the net energy gain reached 4010 MJ/t VS and 3298 MJ/t VS, referring to energy recovery values of 27% and 22%, respectively. The promising energy recovery values achieved by O_2-NBW can be attributed to its 47% higher methane production compared to the control [38].

Liu et al. [145] conducted an energy assessment in a study focused on enhancing CO_2 and H_2 conversion by methanogens, as the limited hydrogen gas–liquid mass transfer often can be a bottleneck in the conversion process. In the two reactors, the CO_2 and H_2 were transferred from the headspace to the reactor broth via continuous recirculation. The two bioreactors were loaded with a micro-nano sparger and a regular micro-sparger. The results showed that the micro-nano sparger generated 26% more methane than the regular micro-sparger. For calculating the energy consumed by the AD system, Liu et al. [145] utilized Eq. (6), while the energy product ratio was calculated using Eq. (7).

$$E_{consumed} = E_{pump} + E_{Stirrer} \quad (6)$$

$$R = \frac{Y_{Methane}}{E_{Consumed}} \quad (7)$$

whereas $E_{consumed}$, refers to the total energy consumed by the AD system. E_{pump} and $E_{stirrer}$ represent the energy consumed by the pump and stirrer, respectively. R and $Y_{Methane}$ refer to the energy-product ratio and cumulative methane production, respectively. Liu et al. [145] reported that the micro-nano sparger consumed 4% and 14% higher pumping and stirring energy, respectively, than the regular sparger. Importantly, the energy-product ratio was 24% higher in the micro-nano sparger than the regular sparger, confirming the high potential for technology scaling up. Noteworthy, this study was conducted at a 500 mL working volume. Therefore, these findings require validation at the pilot scale before application in the industrial-scale facilities, which involve complex and dynamic conditions that must be taken into account.

Future Prospects

Considering that incorporating NBs in AD is still a relatively new area of research, it has considerable potential for further development. To our knowledge, all previous studies investigating the performance and efficiency gains of NBW were performed in batch AD reactors. To the best of the authors' knowledge, no systematic studies have examined the influence of NBW on AD using continuous reactors or pilot-scale reactors, highlighting a significant knowledge gap regarding scale-up challenges. Further studies with continuous and pilot systems are vital for providing a reliable reference for full-scale applications. A thorough investigation of the impact of different operational parameters, such as NBW-to-digester working volume ratio and feedstock type, on NBs-incorporated AD processes and the interaction between these parameters is essential. Attention should be paid to the distribution and injection points of NBs or NBW in the anaerobic digester. Supplementing NBs into anaerobic digesters via multiple ports might be an effective option for ensuring uniform distribution of NBs in the digester. Also, the impact of digester mixing speed should be investigated, as it has a significant effect on digester performance [146, 147]. Moreover, validating the NBW-incorporated AD systems would require developing process control techniques for troubleshooting and optimizing system performance.

Tailoring NBs characteristics, including gas type, bubble size and concentration, to optimize their effects on AD processes is critical. It remains challenging to develop reliable and practical methods for characterizing these features. New horizons must be explored for the NBs generation tools. Lower generation cost and higher NBs stability should be the primary criteria. In addition, a comprehensive techno-economic assessment of NBs integration into AD systems remains absent and is essential to support industrial application. Moreover, developing mathematical models for predicting NBs' impact on AD and proposing the most optimal NBs for each feedstock type can be useful. NB-containing systems have been modeled in previous studies in other fields. In the field of environmental engineering, ozone NBs were utilized for the remediation of trichloroethene contamination in sands, whereas a mathematical model was developed and could appropriately describe the remediation process [148]. Furthermore, considering the removal of butylated hydroxytoluene from water by ozone NBs, Klammler et al. [149] presented a model that included the generation, stability, and treatment reaction mechanisms of NBs. When provided with the reactor size, type of NBs generator, and the gas supply flow, the model can recommend design guidelines. Given the prolonged operation period of the AD process, NBs residence time, which was reported to range from days to months, and its relation to gas type warrant investigation. There is a lack of knowledge in the operational risks (e.g., process instability, inhibition) with NBs supplementation into AD systems. Therefore, long-term operations are highly recommended for monitoring the performance and stability of supplemented NBs. In addition to process optimization, research focus should be directed towards understanding the underlying mechanism of NBs impact on AD. Future research should reveal further approaches for integrating NBs with other process intensification techniques, in such a way that achieves the highest possible synergistic effect. Additionally, the potential role of NBs in enhancing the sludge thickening process warrants further investigation. Improved stability and faster settling rates can be the main outcomes.

Future research should also consider exploring the potential negative effects that could arise from the use of NBs in AD systems. Such negative effects may include instability or inhibition of key microbial consortia, which the high interfacial energy and oxidative nature of some gas types, such as O_2 and O_3 could potentially cause. Another possible cause of the microbial instability is the uneven distribution of NBs within the AD system. Additionally, the long residence time of NBs in the AD system, ranging from days to month [40], may lead to increased microbial stress. Moreover, the extended residence time of NBs in the AD system may disrupt the flocculation process implemented after AD for solid–liquid separation and sludge handling.

Conclusions

Incorporating NBs technology in AD is a rapidly emerging research topic. Numerous previous studies have confirmed the significant positive impact of NBW supplementation in anaerobic digesters. NBs affect all AD stages, hydrolysis, acidogenesis/acetogenesis and methanogenesis. Higher ATP generation, more efficient substrate degradation, microbial community diversification, boosted enzymatic activity, improved nutrients transfer, and enhanced electron transfer rates are the key outcomes of applying NBs in the AD process. The impact of NBs stems from their unique features: the collision of NBs produces reactive radicals, and their distinctive physical properties enable them to disrupt the water structure, thereby enhancing water mobility and reducing viscosity. NBW volume to working volume ratio of 1:2 is the most widely adopted ratio in published reports. However, there is no consensus among studies regarding the optimal gas type. Integrating NBs technology with other AD process intensification techniques improved both methane production yields and the tolerance of inhibitors. Although energy consumption associated with applying NBs in AD

systems appears to be well offset by the extra methane, this conclusion is derived from batch tests and requires further validation.

Author Contributions **Alsayed Mostafa:** Conceptualization, Writing – original draft, Methodology, Investigation, Formal analysis, Visualization; **Ahmed Elsayed:** Writing – original draft, Methodology, Investigation, Formal analysis; **Rasha Maal-Bared:** Writing – review & editing, Methodology, Investigation; **Elsayed Elbeshbishy:** Writing – review & editing, Methodology, Supervision, Investigation; **Bipro Ranjan Dhar:** Writing – review & editing, Resources, Supervision, Project administration, Methodology, Investigation, Funding acquisition.

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Data Availability Data will be made available on request.

Declarations

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Authors and Affiliations

Alsayed Mostafa[1] · Ahmed Elsayed[2] · Rasha Maal-Bared[3] · Elsayed Elbeshbishy[2] · Bipro Ranjan Dhar[1]

✉ Bipro Ranjan Dhar
bipro@ualberta.ca

[1] Department of Civil and Environmental Engineering, University of Alberta, 9211-116, Street NW, Edmonton, AB T6G 1H9, Canada

[2] Department of Civil Engineering, Toronto Metropolitan University, Toronto, ON M5B 2K3, Canada

[3] CDM Smith Canada, Metrotower Office Complex, 4710 Kingsway #1001,, Burnaby, BC V5H 4M2, Canada